

Exam Date & Time: 16-Nov-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

Nanotechnology and Applications [PHY_4345]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Derive an expression for density-of-states function using free electron theory of metals. (5)
 - A)
 - B) Describe the nanoscale materials and their classification. (3)
 - C) Are carbon nanotubes 1D, 2D, or 3D systems? justify your answer. (2)
- 2) Sol gel method is a more significant method to produce nano metal oxide materials, show it with neat diagram & explanations. (5)
 - A)
 - B) Explain differences between laser ablation and molecular beam epitaxy techniques in terms of energy input, deposition control, and scalability. (3)
 - C) Predict what happens to particle size and surface area if the ball milling process duration is increased. Justify your answer briefly (2)
- 3) Apply the principles of optical spectroscopy on UV-Visible absorption and interpret the band gap of the material with neat diagram. (5)
 - A)
 - B) Apply principle of photoluminescence spectroscopy to find suitability of materials for LED applications. (5)
- 4) Apply the synthesis techniques of carbon nanotubes (CNTs) to propose a suitable method for developing CNT-based sensors. (5)
 - A)
 - B) Relate the properties, and applications of nanocomposite metal-matrix fillers. (5)
- 5) Using your understanding of photo-catalysis, propose a nanomaterial-based method for wastewater treatment and assess how nano-scale properties improve catalytic efficiency. (5)

- A)
- B) Identify how nanotechnology can be used to improve targeted drug delivery efficiency. (5)

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Event Name		Subject	
058-END SEMSTER-ODD-2025		PHY_4345:Nanotechnology and Applications	
#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	10
Total			50

* NA - CO Not Selected

Event Name		Subject	
058-END SEMSTER-ODD-2025		PHY_4345:Nanotechnology and Applications	
#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	2	6(12%)
3	Application	10	44(88%)
4	Analysis	0	0(0%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		12	50 (100%)